



BUILD A VIRTUAL PRIVATE CLOUD

By Lenah Mbatha | June 2026

Build a Virtual Private Cloud (VPC)

The screenshot shows the AWS Management Console interface for creating a new VPC. The breadcrumb navigation at the top reads 'VPC > Your VPCs > Create VPC'. The main heading is 'Create VPC' with an 'Info' link. Below this, a descriptive sentence states: 'A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.'

The 'VPC settings' section contains the following fields and options:

- Resources to create:** Two radio button options: 'VPC only' (selected) and 'VPC and more'.
- Name tag - optional:** A text input field with a placeholder 'Network VPC'. A note above it says: 'Creates a tag with a key of "Name" and a value that you specify.'
- IPv4 CIDR block:** A section with two radio button options: 'IPv4 CIDR manual input' (selected) and 'IPAM-allocated IPv4 CIDR block'. Below these is a text input field containing '10.0.0.0/16'. A note below the field states: 'CIDR block size must be between /16 and /28.'
- IPv6 CIDR block:** A section with four radio button options: 'No IPv6 CIDR block' (selected), 'IPAM-allocated IPv6 CIDR block', 'Amazon-provided IPv6 CIDR block', and 'IPv6 CIDR owned by me'.

At the bottom left, there is a 'Tenancy' section with an 'Info' link.

Introducing Today's Project!

In this project, I will demonstrate how to build a virtual private cloud

What is Amazon VPC?

Amazon VPC is its a Virtual Private Cloud and it is useful because it helps you organize you AWS resources in a secure manner based on their similarities and security permissions

In today's project, I used Amazon VPC to create a subnet and an Internet Gateway

Personal reflection

This project took me 2 hours

One thing I didn't expect in this project was the complexity of the CLI commands

Virtual Private Clouds (VPCs)

What I did in this step

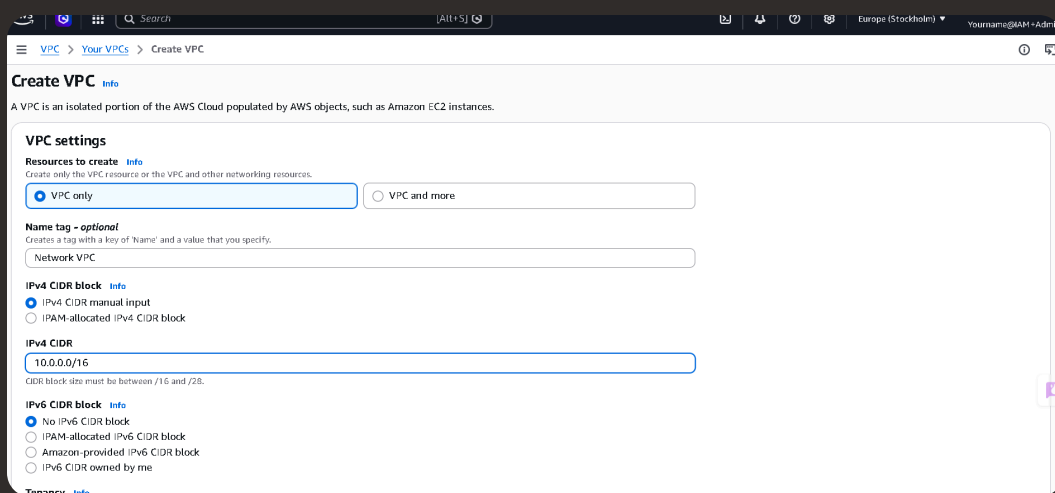
In this step, I will access the vpc in my IAM user cocole and assign ip CIDR Adresses

How VPCs work

VPCs are Virtual Private Clouds that isolate sections in the cloud, making the users organize their cloud resources

Why there is a default VPC in AWS accounts

There was already a default VPC in my account ever since my AWS account was created. This is why users could launch EC2 instances since that day they opened an AWS account. We can create other EC2 instances and customize them, but the default one is a handy starting point, especially for beginners.



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The 'VPC settings' section contains the following options:

- Resources to create:** Two radio buttons are present. 'VPC only' is selected, while 'VPC and more' is unselected.
- Name tag - optional:** A text input field is provided with the placeholder text 'Network VPC'.
- IPv4 CIDR block:** This section has two radio buttons. 'IPv4 CIDR manual input' is selected, and 'IPAM-allocated IPv4 CIDR block' is unselected. Below this, a text input field contains the value '10.0.0.0/16'. A small note below the field states: 'CIDR block size must be between /16 and /28.'
- IPv6 CIDR block:** This section has four radio buttons. 'No IPv6 CIDR block' is selected. The other three options are 'IPAM-allocated IPv6 CIDR block', 'Amazon-provided IPv6 CIDR block', and 'IPv6 CIDR owned by me', all of which are unselected.

The bottom of the console shows a 'Tenancy' section with an 'Info' link.

Defining IPv4 CIDR blocks

To set up my VPC, I had to define an IPv4 CIDR block, which is ClasslessInter-Domain Routing, which is a way of assigning block of addresses

Subnets

What I did in this step

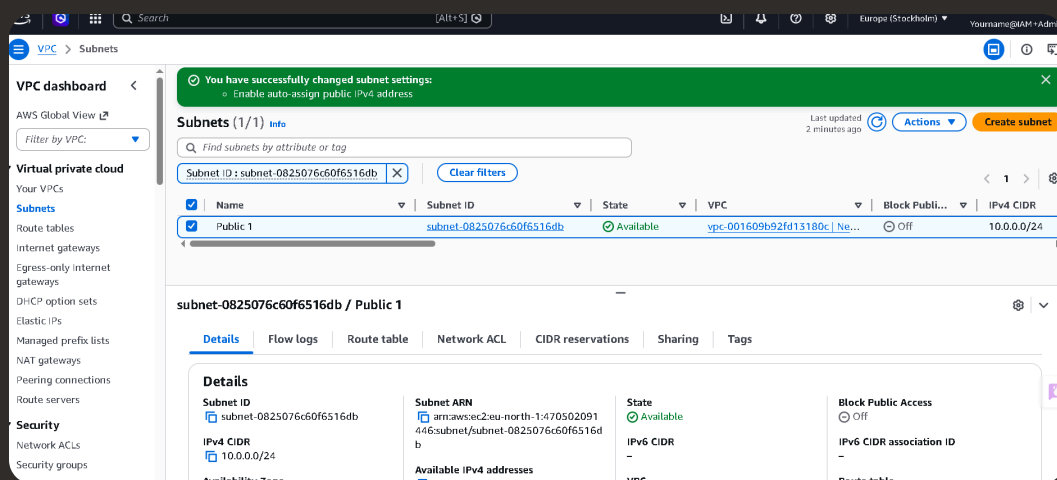
In this step, I will assign a public address that can be used by users to access public resources.

Creating and configuring subnets

Subnets are like neighbourhoods in a private city. You group subnets together based on their similarities and their restrictions. There are already subnets existing in my account, one for every Availability Zone.

Public vs private subnets

The difference between public and private subnets is that public subnets contain resources that can communicate with external networks, while resources in private networks like databases cannot communicate with external networks. For a subnet to be considered public, it has to have an EC2 instance running a public resource like a web user interface.



Auto-assigning public IPv4 addresses

Once I created my subnet, I enabled. auto-assign IPv4 addresses. This setting makes sure that any EC2 instance launched in that subnet will have a public IPv4 address so that there's no need to do it manually again

Internet gateways

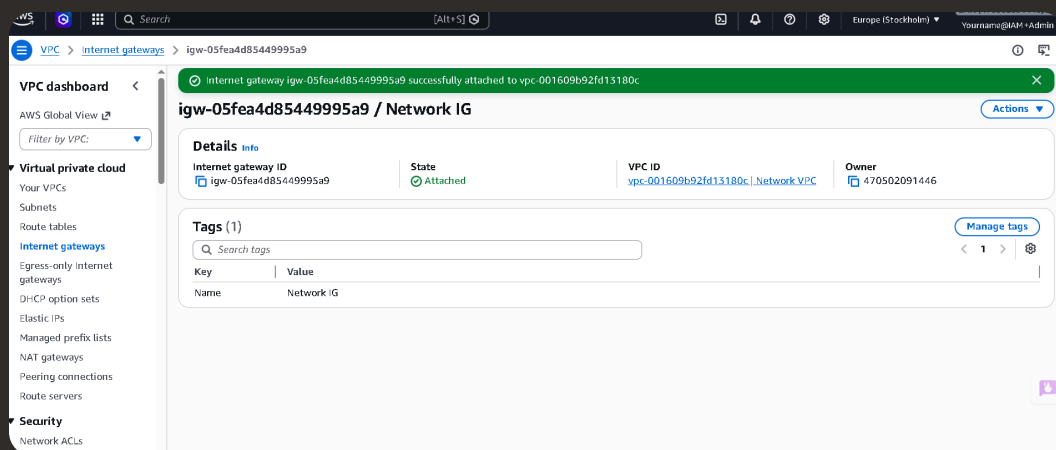
What I did in this step

In this step, I will connect my VPC to an Internet gateway because I want my resources to communicate beyond private space

Setting up internet gateways

Internet gateways are ways that make resources accessible beyond private space

Attaching an internet gateway to a VPC means I am allowing my resources inside the VPC to be accessible to the public. If I missed this step, there will be no communication between my VPC and the Internet



Using the AWS CLI

What I'm doing in this extension

In this project extension, I will launch AWS resources using AWS CLI commands because i want to confirm whether its faster and efficinet that using the AWS Management Console.

Exploring CloudShell and CLI

VPC resources could also be created with CloudShell, which is a way of scripting code instead of creting the resourcesmanually.. CLI is Command line interface which allows you to edit the script for managing your AWS resources

Debugging my setup

To set up a VPC or a subnet, you can use the command (aws ec2 create-subnet --vpc-id vpc-001609b92fd13180c --cidr-block 10.0.1.0/24)



```
CloudShell
eu-north-1 x eu-north-1 x +
~$ aws ec2 create-tags --resources=vpc-001609b92fd13180c --tags Key=Name,Value="NextWork VPC 2"
~$ aws ec2 create-subnet --vpc-id vpc-001609b92fd13180c

aws: [ERROR]: An error occurred (MissingParameter) when calling the CreateSubnet operation: The request must contain at least one of these parameters: cidrBlock or ipv6cidrBlock or ipv6ipamPoolId or ipv6ipamPoolId.
~$ aws ec2 create-subnet --vpc-id vpc-001609b92fd13180c --cidr-block 10.0.0.0/24

usage: aws [options] <command> [<subcommand> ...] [<parameters>]
To see help text, you can run:

aws help
aws <command> help
aws <command> <subcommand> help

aws: [ERROR]: Unknown options: 10.0.0.0/24
~$ aws ec2 create-subnet --vpc-id vpc-001609b92fd13180c --cidr-block 10.0.0.0/25

aws: [ERROR]: An error occurred (InvalidSubnet.Conflict) when calling the CreateSubnet operation: The CIDR '10.0.0.0/25' conflicts with another subnet
~$ aws ec2 create-subnet --vpc-id vpc-001609b92fd13180c --cidr-block 10.0.0.0/24
```

Comparing CloudShell vs AWS Console

Compared to using the AWS Console, an advantage of using commands is its prone to error. An advantage of using the Console iseasy to use for a beginner. Overall, I preferred AWS Console

the 1990s, the number of people in the UK who are aged 65 and over has increased by 1.5 million (1990–1999) and is projected to increase by a further 1.5 million by 2010 (Office of National Statistics 2000). The number of people aged 65 and over is projected to increase by 2.5 million by 2020 (Office of National Statistics 2000).

There is a growing awareness of the need to develop strategies to meet the needs of the ageing population. The Department of Health (1999) has published a strategy for the ageing population, which sets out the government's commitment to improve the health and social care of older people. The strategy is based on the following principles: (1) to improve the health and social care of older people; (2) to ensure that older people are able to live independently; (3) to ensure that older people are able to participate in society; and (4) to ensure that older people are able to live in their own homes.

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